MANAGEMENT CONSULTING CASE STUDY · PILOT

How exposed is IT-BPM work to generative AI, and where should the workforce budget go?

Software developers · QA testers · Customer service · hypothetical 10,000-person client

Redesign work and monitor first, redeploy selectively, and do not mass-reskill

₹17.9 cr
3-year NPV, base case

98%
ROI on present value of costs

Year 2
cumulative payback

4,380
of 7,200 modelled employees covered

Reskilling adds value for developers but delivers about half of what redesign does per employee, and it destroys value in customer service (−₹43k per head). Training the most exposed role first loses ₹8.7 crore.

Source: pilot model; hypothetical 10,000-person client; INR inputs are assumptions

The decision: split a fixed budget across four workforce moves

Questions answered

- Which roles change most
- Reskill, redeploy, hire or redesign
- How many employees, at what cost
- 3-year return and 12-month actions

Term	Meaning
Exposure	Tasks GenAI could affect
Automability	Tasks AI could run with limited oversight
Adoption	What firms actually deploy
Displacement	Jobs actually lost; never inferred from scores

Source: pilot model; hypothetical 10,000-person client; INR inputs are assumptions

Observed data anchor the scores; money inputs are assumptions

Input	Source	Status
Exposure benchmark	ILO WP140 (2025), Table A1, 112 ISCO-08 occupations	Observed
Task statements	O*NET OnLine, 60 tasks, 3 occupations	Observed
Sector size	Indian Labour Statistics 2022-23, QES IT/BPO	Observed; 2.1M-3.8M swings are survey artefacts
Task scores	Claude Sonnet 5.5, single pass, rubric v1.0	LLM output; human validation pending
Wages, costs, adoption, headcount	Analyst assumptions, low/base/high	Assumed

Limits: 3 occupations, one scoring pass, hypothetical client

Customer service is most exposed; developers keep the most human-led work

Occupation	Exposure	Automation	Augmentation	Human judgment	Feasibility	ILO mean
Software developers	0.63	0.39	0.70	0.85	0.80	0.53
QA analysts & testers	0.74	0.56	0.78	0.73	0.88	0.55
Customer service reps	0.85	0.69	0.85	0.60	0.87	0.58

The ranking matches the ILO, which places all three in Gradient 3. Levels differ because the ILO scores task automation potential and ours add a limited-oversight test. Three occupations make this indicative only.

Source: LLM-scored O*NET tasks; ILO WP140 Table A1

Each role splits differently into automate, augment and human-led work

Developers

Coding, testing, documentation and reporting score 4/4 on exposure. Requirements, design reviews and supervision stay human: 51% of task weight is human-led.

QA testers

Test design, defect write-ups, script upkeep and document review score highest. Developer feedback, release standards and site visits stay human: 24%.

Customer service

Record keeping, forms and templated replies are close to fully automatable. Insurance coverage review and complaint resolution need verification: only 5% is purely human-led.

Source: LLM-scored O*NET tasks, importance-weighted

Redesign beats reskilling per employee; redeployment pays but is capped

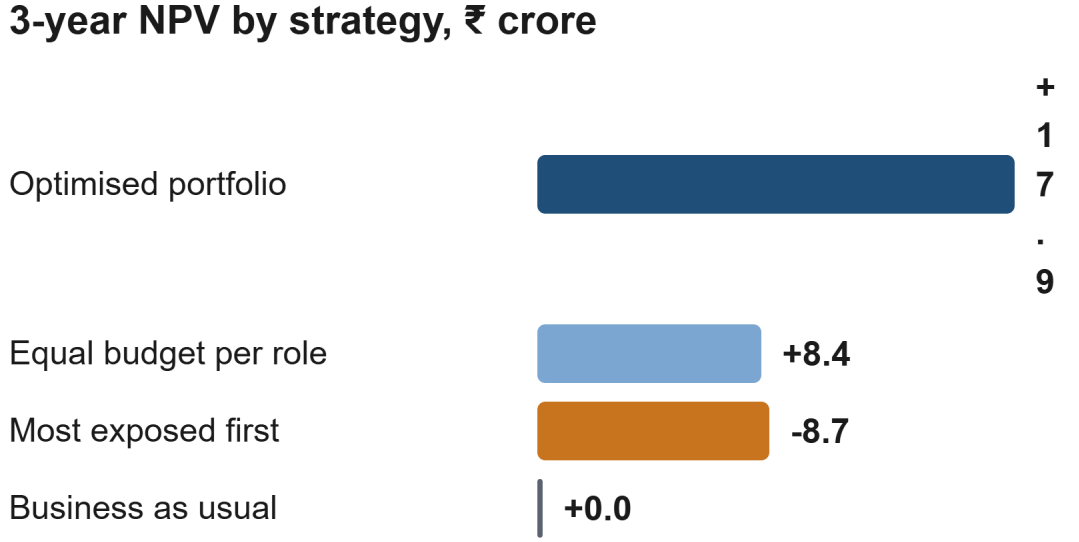
3-year NPV per employee (₹)	Redesign & monitor	Reskill	Redeploy*	Hire specialists
Software developers	+39,000	+19,500	+157,500	−66,300
QA testers	+21,700	−4,400	+99,200	−89,300
Customer service	−5,400	−42,600	+47,600	−125,500

*Redeployment is limited to 6% of each role because it depends on demand in target roles. Discount rate 12%; all unit costs and benefits are assumptions.

Source: pilot model; hypothetical 10,000-person client; INR inputs are assumptions

The optimised portfolio creates ₹17.9 crore; training the most exposed destroys ₹8.7 crore

Role	Allocation
Developers	2,820 redesign, 180 redeploy
QA testers	1,128 redesign, 72 redeploy
Customer service	180 redeploy; 2,640 unchanged



Source: pilot model; hypothetical 10,000-person client; INR inputs are assumptions

Value stays positive in every tested case but depends on adoption and productivity

Driver	NPV range, ₹ cr
AI adoption	6.2 to 30.2
Productivity realisation	6.0 to 30.8
Workforce size	14.3 to 20.7
Time to benefit	13.8 to 20.7
Budget (₹15-40 cr)	14.0 to 17.9

Stable: redesign for developers and QA, redeployment within caps, no layoffs.

Assumption-dependent: reskilling enters only if training cost falls 30% (405 people) or its productivity multiplier reaches 1.7 (248).

Break-even: benefits can fall about 50% before NPV reaches zero.

Source: pilot model; hypothetical 10,000-person client; INR inputs are assumptions

Next 12 months: pilot, measure, then scale at a decision gate

Months 0-3

Replace assumed wages and headcount with client data. Pilot AI tooling in two developer and QA teams. Human-label the 24-task validation sample.

Months 3-6

Scale tooling where adoption tracks the base path. Track adoption, cycle time, rework and defects. Open 180, 72 and 180 redeployment slots.

Months 6-12

Gate: continue if adoption and benefits reach at least half of plan. Re-test reskilling with measured costs.

Decisions required: approve a ₹22 crore 3-year envelope, cap redeployment at 6%, and fix the month-12 gate.

Source: pilot model; hypothetical 10,000-person client; INR inputs are assumptions